

Appendix

X Additional Results

A. In detail Results on Sparse ResNet18

Results on CIFAR-10: As shown in Table II, ResNet18 with KD consistently maintains high accuracy across sparsity levels. At the extreme compression of 98.41%, the model achieves 90.79%, while at 90.00% sparsity accuracy reaches 94.97%. Even at the lowest sparsity (50.46%), KD preserves a strong accuracy of 96.08%. These results illustrate that the proposed KD-guided pruning framework allows the student network to retain remarkable performance despite aggressive parameter reduction, with $T = 3$ playing a key role in stabilizing performance under extreme sparsity.

Results on CIFAR-100: On the more challenging CIFAR-100 dataset, KD enables the student model to maintain competitive accuracy across a wide range of sparsity levels. At 98.01% sparsity, the model still achieves 66.32%, and performance steadily improves as sparsity decreases: 75.34% at 92.53%, 77.40% at 90.55%, and 79.48% at 80.60%. The best accuracy of 81.01% is observed at 50.75% sparsity. This trend demonstrates the resilience of the proposed framework, where even under extreme compression the student model retains strong generalization, and accuracy progressively recovers as more parameters are preserved.

Results on TinyImageNet: The effectiveness of the framework is further validated on TinyImageNet, which poses a more complex recognition task. At extreme sparsity (97.56%), KD achieves 50.64% accuracy, showing the models ability to withstand heavy pruning. As sparsity is reduced, performance improves notably, reaching 58.79% at 93.11%. At moderate sparsity (89.22%), accuracy stabilizes at 52.36%, while at lower sparsity levels KD provides consistent gains, with 56.54% at 79.31% and 58.49% at 74.35%. The best accuracy of 59.29% is obtained at 50.02% sparsity. These results confirm that the KD-guided pruning framework generalizes effectively to large-scale datasets, ensuring stable performance across both extreme and moderate compression regimes.

Overall, the results across CIFAR-10, CIFAR-100, and TinyImageNet demonstrate the robustness and adaptability of the proposed KD-guided pruning framework. By leveraging the CA-KLD objective with $T = 3$, the method consistently recovers or even improves student model accuracy under severe sparsity, establishing its effectiveness as a general solution for compressing deep networks while preserving discriminative power.

B. Results On Sparse Resnet34

Results On CIFAR-10: Table VIII presents the results for ResNet34 across sparsity levels from 50.47% to 98.44%. At the highest sparsity of 98.44%, KD with $T = 3$ achieves 92.35%, a 6.13 percentage-point improvement over the non-KD accuracy of 86.22%. At 90.02% sparsity, KD improves performance from 94.93% (non-KD) to 95.71% ($T = 3$). At 50.47% sparsity, the model retains high accuracy with 96.52% even without KD, while KD retraining sustains competitive

TABLE VI: Accuracy Comparison for Sparse ResNet18 with EPG

Dataset	Sparsity (%)	Our Method		Liao et al. [20]
		Acc. $T=3$ (%)	Acc. $T=5$ (%)	Acc. (%)
CIFAR-10	50.00	96.08	96.03	92.56
	75.00	95.62	95.60	93.00
	93.92/93.8	94.28	94.40	92.93
	98.41/98.4	90.79	90.66	93.12
TinyImageNet	50.02/50.00	59.29	60.93	41.70
	74.35/75.00	57.42	58.58	41.24
	93.92/93.8	53.79	52.11	41.86
	98.41/98.4	50.64	50.07	37.44

performance. Interestingly, at 80.02% sparsity, both $T = 3$ and $T = 5$ achieve 96.12%, slightly outperforming the non-KD accuracy of 96.16%. This suggests smoother logits (higher T) may provide subtle gains in denser models. Overall, ResNet34 consistently outperforms ResNet18 under extreme pruning, achieving 92.35% at 98.44% sparsity (KD, $T = 3$) versus 90.79% for ResNet18, highlighting the advantage of deeper student networks in retaining performance under high sparsity conditions.

Results On CIFAR-100: Table VII compares the performance of the deeper ResNet34 student model under the same teacher and conditions. At 98.22% sparsity, KD yields accuracies of 72.38% ($T = 3$) and 72.79% ($T = 5$), improving significantly over the 66.14% baseline. At 90.75% sparsity, performance rises to 81.09% ($T = 3$) and 81.44% ($T = 5$), versus 77.89% without KD. With sparsity at 80.77%, KD achieves 82.21% ($T = 3$) and 81.91% ($T = 5$), outperforming the baseline of 81.09%. At 76.56% sparsity, accuracies of 82.21% ($T = 3$) and 82.25% ($T = 5$) are recorded, compared to 81.70% without KD. Interestingly, at the lowest sparsity level (50.86%), KD slightly underperforms the baseline of 83.18%, with 83.12% ($T = 3$) and 82.83% ($T = 5$). These results highlight ResNet34's stronger baseline and consistent performance with KD, especially under higher sparsity, while maintaining competitive accuracy as sparsity decreases.

XI Additional Comparison Result

A. Comparison with EGP.

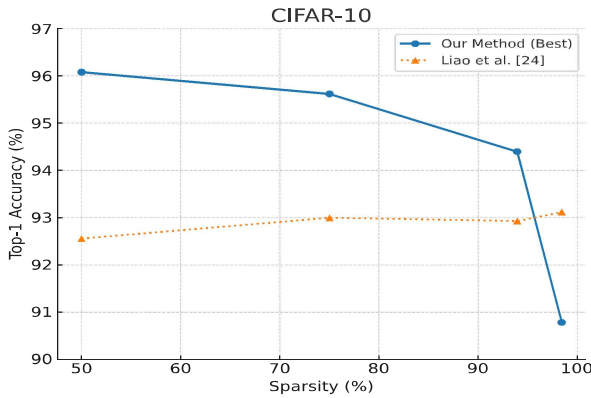
We also compare our method with the entropy-guided pruning(EGP) approach by Liao et al. [20] across various sparsity levels for ResNet18 on CIFAR-10 and TinyImageNet, as shown in Table VI. Our method uses KD configurations with $T = 3$ and $T = 5$, both with $\alpha = 0.7$, and reports the best accuracy between the two. and line plot visualization in Figure 7 Our teacher-guided pruning approach generally outperforms Liao et al. [20] across most sparsity levels (Table VI). On CIFAR-10, at 50.00% sparsity, our method achieves 96.08% accuracy ($T = 3$), surpassing Liao et al.'s 92.56% by 3.52 percentage points. This trend continues at 75.00% sparsity (95.62% vs. 93.00%) and 93.92% sparsity (94.40% vs. 92.93% at 93.8% sparsity for Liao et al. [20]). However, at the highest sparsity of 98.41%, Liao et al.'s method achieves 93.12%

TABLE VII: Performance of the student model (ResNet34) distilled from the teacher model (ResNet50) on CIFAR-100 using our method. Results are reported across different sparsity levels in terms of accuracy without knowledge distillation (w/o KD) and with KD at temperatures $T = 3$ and $T = 5$ with $\alpha = 0.7$.

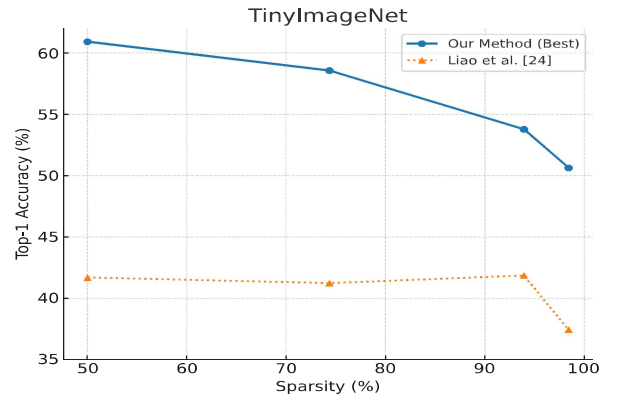
Sparsity (%)	Accuracy w/o KD (%)	Acc (KD, T=3, $\alpha=0.7$) (%)	Acc (KD, T=5, $\alpha=0.7$) (%)	# Epoch w/o KD (Retrain)	# Epoch KD T=3 (Retrain)	# Epoch KD T=5 (Retrain)
98.22	66.14	72.38	72.79	22	38	39
93.74	75.35	79.54	79.78	10	34	22
90.75	77.89	81.09	81.44	8	11	30
80.77	81.09	82.21	81.91	13	28	14
75.85	81.70	82.41	82.25	9	11	10
50.86	83.18	83.12	82.83	14	11	7

TABLE VIII: Performance of the sparse student model (ResNet34) distilled from the teacher model (ResNet50) on CIFAR-10 using our method. Results are reported at various sparsity levels in terms of accuracy without knowledge distillation (w/o KD) and with KD at temperatures $T = 3$ and $T = 5$ with $\alpha = 0.7$.

Sparsity (%)	Accuracy w/o KD (%)	Acc (KD, T=3, $\alpha=0.7$) (%)	Acc (KD, T=5, $\alpha=0.7$) (%)	# Epoch w/o KD (Retrain)	# Epoch KD T=3 (Retrain)	# Epoch KD T=5 (Retrain)
98.44	86.22	92.35	91.57	16	44	37
93.94	94.07	95.16	95.38	34	20	22
90.02	94.93	95.71	95.67	20	11	18
80.02	96.03	96.08	96.12	15	9	28
75.01	96.16	96.13	96.14	11	13	11
50.47	96.52	96.51	96.49	13	9	17



(a) Cifar10



(b) Tiny ImageNet

Fig. 7: Top-1 accuracy vs. sparsity for ResNet18 on CIFAR-10 and TinyImageNet. Results are shown for our method (solid blue line) and the method by Liao et al. [20] (dotted orange line) across varying sparsity levels. Accuracy is reported at different sparsity points, with detailed values provided in Table VI

accuracy (at 98.4% sparsity), outperforming our 90.79% by 2.33 percentage points, possibly due to their entropy-guided strategy better preserving critical neurons at extreme sparsity. On TinyImageNet, our method consistently excels: at 50.02% sparsity, we achieve 60.93% accuracy ($T = 5$) compared to 41.70% for Liao et al. (at 50.00% sparsity), a 19.23 percentage-point improvement. This gap persists at 74.55% sparsity (76.58% vs. 41.24% at 75.00% sparsity), 93.92% sparsity (53.79% vs. 41.86% at 93.8% sparsity), and 98.41% sparsity (50.64% vs. 37.44% at 98.4% sparsity). The superior performance on TinyImageNet highlights the effectiveness of

our CA-KLD-based KD (Section III-B) in transferring fine-grained knowledge, enabling better generalization on more complex datasets with higher class diversity.